

Transforming semantic embedding functions for agentic traces

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The problem

Modern generative systems produce logs of their actions and decisions—referred to as an *agentic trace*—while processing information. Traces often implicitly or explicitly encode information such as the particular type of tool that was used, the harness, etc. that may or may not be useful for determining if the trace produced a useful output. Further, the signal from using an off-the-shelf embedding function to encode trace similarity is often dominated by information that is potentially irrelevant for correctness, such as the harness or model used: in our corpus, nearest neighbors of raw trace embeddings are same-system 68–83% of the time, same-scaffold at $5\times$ chance, and the exact agent configuration is retrievable at 0.82 vs. 0.43 chance even among same-model, same-harness siblings—while agreement with outcome sits at chance (AUC 0.52).

The goal of this work is simple: **develop a method for embedding agentic traces that better captures similarity of the work performed.** In developing such a method, we have identified six desirable properties of an agentic trace embedding, organized into three groups:

Faithful to content — what happened should drive similarity.

- **Task fidelity:** traces of different systems working the same problem should be similar. *Measured by* cross-system same-task nearest-neighbor retrieval.
- **Behavioral fidelity:** among traces of the same problem, similar approaches and results should be similar. *Measured by* within-task coherence of same-outcome traces (AUC of same- vs. different-outcome distances).

Invariant to authorship — who produced it should not.

- **Harness invariance:** scaffold formatting and tool idiom should not drive similarity. *Measured by* within-task same-scaffold retrieval vs. chance.
- **Model-family invariance:** the vendor/lineage of the underlying LLM should not drive similarity. *Measured by* within-task same-family retrieval vs. chance.
- **Identity invariance:** the exact system configuration (version, prompts, settings) should not be recoverable. *Measured by* exact-submission retrieval among same-model, same-harness siblings vs. chance.

Reliable — similarity should be a measurement, not noise.

- **Stability:** the embedding should be stable under agent stochasticity, judge resampling, and embedder choice. *Measured by* replicate-run and judge-resample agreement, per trace and aggregated over instances.

qubric

Our method for embedding agentic traces is based on a query-specific rubric. In particular, for each task instance an LLM (Model 1) reads the public problem statement and writes a query-specific rubric comprised of 1) understanding, 2) localization, 3) reproduction, 4) editing, 5) verification, and 6) final state. Given a trace corresponding to the task, a judge LLM (Model 2) extracts a short factual description per section. The short descriptions are then embedded with an off-the-shelf embedding model, centered against the cross-system median per instance, and concatenated. This approach enforces the equivalence relations the space should respect by respecting the rubric and using a fixed judge LLM as the extractor.

Evidence

We evaluate qubric on 107 system configurations from the SWE-bench Verified leaderboard (a 500-task benchmark; qubric judging is on a 20-task panel, with the 150-task scale-up in progress). We use gpt-5.4-mini for Model 1 and gpt-5.4-mini for Model 2 in the primary configuration; robustness to Model 2 is reported below. We compare qubric along the six axes described above against four baselines:

- **raw trace:** embed the first/last 8K tokens of the rendered trace directly—the off-the-shelf default;
- **free-form summary:** Model 2 summarizes the trace with no rubric—tests whether structure matters;
- **generic rubric:** the same six sections, fixed across instances—tests whether query-specificity matters;
- **verdict questions:** the rubric phrased as evaluative questions with short answers—tests descriptions vs. verdicts;

across **seven off-the-shelf embedding functions** spanning three families and two orders of magnitude of size. Per-pillar results for the baselines are shown in Fig. 2.

Qubric produces the same profile shift under all seven embedders (Fig. 1): *task fidelity* is retained (~ 0.95) and *behavioral fidelity* retained or improved, while every authorship axis collapses to $\approx$ chance—*harness* $0.31 \rightarrow 0.08$, *model family* $0.20 \rightarrow 0.12$, *identity* $0.83 \rightarrow 0.09$. The properties are separately measurable (double dissociations across six representations), so these are genuinely different axes being moved, not one metric relabeled.

The instance-conditioning is the active ingredient: the construction ranking is strict under both readouts (qubric > verdict questions > generic rubric > free-form summary > raw), reproduces under a local embedder, and no single

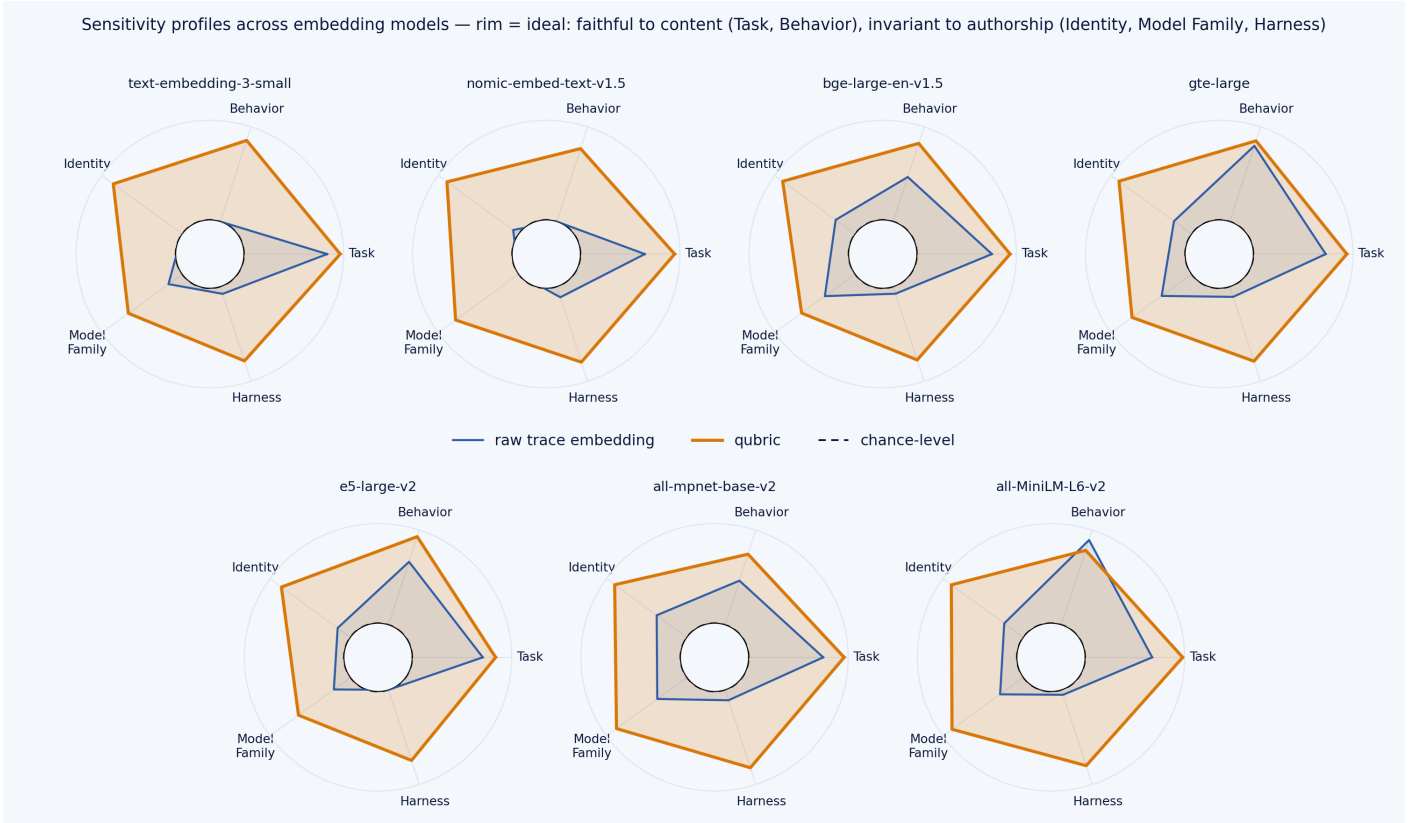


Figure 1: Sensitivity radars across seven embedders. Outside = desirable. Raw embeddings (navy) are authorship-dominated; qubric (amber) collapses Identity / Model family / Harness to chance while holding Task and Behavior.

Pillar desirability across representations (cell = raw metric; blue = ideal, red = undesirable)

	Task	Behavior	Identity	Model Family	Harness
raw	0.83	-0.01	0.92	0.43	0.32
head-only	0.78	-0.02	0.83	0.42	0.24
tail-only	0.88	-0.01	0.93	0.47	0.38
centered	0.37	0.03	0.92	0.41	0.32
free-form judge	0.53	0.02	0.79	0.38	0.14
qubric	0.95	0.02	0.52	0.36	0.08

desirability (1 = ideal for this pillar)

Figure 2: Baselines by pillar: each cell is the raw metric for that axis (columns) under that representation (rows); color is desirability (blue = ideal). Raw-trace variants (head/tail/centered) read authorship; qubric is the only row that is simultaneously task-faithful and authorship-invariant.

rubric section carries the effect (best single section 0.091 vs. 0.078 for the union). Judge choice degrades gracefully: an open frontier judge (DeepSeek-v3.1, 0.083/0.079) matches the closed reference (gpt-5.4-mini, 0.078/0.078) within panel resolution, and a weak judge *with* a qubric outperforms a strong judge without one.

Stability: two judges agree on a single trace only 12–28% of the time, but aggregating ~ 20 instances drives system-level agreement to 0.82–0.95; replicate runs of the same agent

retrieve each other at $6\times$ chance. Per-trace noisy, per-agent reliable—and per-agent is what evaluation uses.

Query-efficient benchmarking

Qubric representations enter the DKPS/PKPS perspective space; a new agent’s full-benchmark resolve rate is predicted from a few probe traces against references’ cached coverage, blended with the sample score. Evaluation is leave-one-LLM-out throughout (naive CV inflates accuracy $\sim 25\%$ via model-family leakage).

probes m	sample score	qubric pipeline	+selection
1	0.44	0.095	—
5	0.16	0.074	0.057
20	0.06	0.048–0.053	0.053

Table 1: MAE predicting true resolve rates (107 systems). One probe trace ≈ 8 –10 scored runs.

One trace carries far more than one bit: the strongest correctness-only competitor (CAPA-style error overlap) needs 15–20 *scored* probes to match one *trace* (0.37 MAE at $m=1$), and trails the ensemble at every budget. The reference pool, not probe count, is the binding resource: error falls $0.145 \rightarrow 0.078$ as the pool grows $5 \rightarrow 96$, still declining (Fig. 3). Every layer is ablated; CV-inside-selection and PCA-composed-with-selection are documented overfitting traps.



Figure 3: QUENCH: error predicting true resolve rates vs. probe budget m (left) and vs. reference-pool size n (right); leave-one-LLM-out, 107 systems.

Remaining tasks

1. Sensitivity to the choice of Model 1 and Model 2.
2. An additional benchmark, ideally Terminal-Bench (142 systems, 89 tasks, ≥ 5 trials per task, public trajectories).
3. A descriptive figure: trace anatomy and the qubric pipeline.
4. Actual dollar savings from probe-based evaluation vs. full benchmark runs.
5. Literature review.
6. Another downstream task? (Routing? Just needs to beat using the off-the-shelf embedding for this paper.)

Appendix: a worked example (django__django-11099)

A. Example trace (excerpt)

The opening and closing of a sweagent + GPT-4 trace on this instance. Note how much of the text is harness syntax, state dumps, and format echoes rather than content.

```
swe_main
First, we should locate the files that contain the 'ASCIIUsernameValidator' and 'UnicodeUsernameValidator' classes to understand where the
  regex patterns are defined. Since the issue mentions that these validators are in 'contrib.auth.validators', we can start by searching
  for files named 'validators.py' within the 'contrib/auth' directory.

action: find_file "validators.py" contrib/auth

Directory contrib/auth not found

First, we should locate the files that contain the 'ASCIIUsernameValidator' and 'UnicodeUsernameValidator' classes to understand where the
  regex patterns are defined. [...]

'''
find_file "validators.py" contrib/auth
'''
{"open_file": "n/a", "working_dir": "/django__django"}

It seems there was a misunderstanding with the directory structure. Let's try to find the 'validators.py' file without specifying the
  directory [...]

action: find_file "validators.py"

Found 3 matches for "validators.py" in /django__django:
/django__django/django/core/validators.py
/django__django/django/contrib/auth/validators.py
/django__django/django/contrib/postgres/validators.py

[... ~40K tokens omitted ...]

(Open file: /django__django/regex_test.py)
(Current directory: /django__django)
bash-$
primary
role: assistant
The output indicates that both the 'ASCIIUsernameValidator' and 'UnicodeUsernameValidator' regex patterns correctly rejected the username
  with a trailing newline. This confirms that the changes made to the regex patterns are effective.

Since the issue has been addressed and the fix has been verified with a test script, we can now submit our changes to the code base.

The output indicates that both the 'ASCIIUsernameValidator' and 'UnicodeUsernameValidator' regex patterns correctly rejected the username
  with a trailing newline. [...]

'''
submit
'''
action: submit
```

B. Query-specific rubric generation (Model 1)

The problem statement for this instance:

```
UsernameValidator allows trailing newline in usernames
Description

ASCIIUsernameValidator and UnicodeUsernameValidator use the regex
r'^[\w.@+-]+$'
The intent is to only allow alphanumeric characters as well as ., @, +, and -. However, a little known quirk of Python regexes is that $ will
  also match a trailing newline. Therefore, the user name validators will accept usernames which end with a newline. You can avoid this
  behavior by instead using \A and \Z to terminate regexes. For example, the validator regex could be changed to
r'\A[\w.@+-]+\Z'
in order to reject usernames that end with a newline.
I am not sure how to officially post a patch, but the required change is trivial - using the regex above in the two validators in contrib.
  auth.validators.
```

The rubric-generation prompt, with the problem statement above substituted for {problem}:

```
Here is a repository issue that a software-engineering agent will be asked to fix:

{problem}

Write an instance-specific rubric for judging an agent's trace on THIS issue:
for each section below, one sentence saying what specifically to look for in
this instance (name the likely relevant modules/files/tests if inferable).

Sections: understanding, localization, reproduction, editing, verification, final_state.
```

Respond with ONLY a JSON object with those keys, each a one-sentence string.

The resulting query-specific rubric:

```
{
  "understanding": "Look for the agent correctly identifying that the bug is in Django's auth username validators accepting usernames with a trailing newline because the regex uses '$', and that the fix should tighten the anchors rather than change the allowed character class.",
  "localization": "Look for the agent localizing the problem to 'contrib/auth/validators.py' (likely 'ASCIIUsernameValidator' and 'UnicodeUsernameValidator') and, if they inspect tests, to any auth validator tests under 'tests/auth_tests/' or similar.",
  "reproduction": "Look for a reproduction that demonstrates a username like 'alice\n' passes validation with the current regex, ideally via the validator classes or a small shell/test snippet showing '$' matches before a final newline.",
  "editing": "Look for the agent changing the validator regexes from 'r'^[\\w.@+-]+$' to 'r'^A[\\w.@+-]+Z$' in both validators, without altering the intended character whitelist or unrelated auth logic.",
  "verification": "Look for the agent adding or updating tests that assert usernames ending in newline are rejected by both ASCII and Unicode validators, and running the relevant auth test module or focused test subset to confirm the fix.",
  "final_state": "Look for the final state to include only the targeted regex anchor fix and passing/updated tests, with no regressions or broader API changes in auth username validation."
}
```

The judge (Model 2) reads the full trace through this rubric and emits one short factual description per section; these six texts are what gets embedded:

```
{
  "understanding": "The agent correctly identified that Django's username validators accept trailing newlines due to regex anchors ^ and $, and decided to fix it by tightening the anchors to A and Z without changing the character class.",
  "localization": "The agent localized the problem to django/contrib/auth/validators.py, specifically the ASCIIUsernameValidator and UnicodeUsernameValidator classes, by searching for and opening the file. It did not inspect any auth validator tests.",
  "reproduction": "The agent created test scripts to reproduce the issue, including one with a username like 'testuser\n'. However, Django-based tests failed due to settings configuration; it successfully used a regex-only script to demonstrate the problem.",
  "editing": "The agent edited both ASCIIUsernameValidator and UnicodeUsernameValidator in validators.py, changing the regex patterns from r'^[\\w.@+-]+$' to r'^A[\\w.@+-]+Z$', keeping the character class intact.",
  "verification": "The agent created custom test scripts, including regex_test.py, which showed that usernames with trailing newlines were rejected. It did not add or update existing Django tests or run the auth test module.",
  "final_state": "The final state includes only the regex anchor fix in validators.py and the addition of test script files. No regressions or broader API changes were made, and the fix was verified via custom tests."
}
```